EECE 460 Project Report

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***Abstract*—This paper presents the modeling and control of a one-degree-of-freedom (1-DOF) copter platform. The objective is to stabilize the pitch angle while satisfying strict performance constraints, including a settling time below 1.5 s, an overshoot under 15%, and a steady-state error less tha 3°. A physics-based dynamic model is first derived and then refined through experimental system identification to correct for parameter mismatch. Using the validated model, a proportional–integral–derivative (PID) controller and a Lag-Lead compensator are designed employing both root-locus and frequency-response methods. Simulation results demonstrate that the final controller meets all performance specifications, confirming the suitability of the proposed design for hardware implementation.**

***Index Terms*—1-DOF copter, system identification, PID control, Lag-Lead, root locus, frequency response.**

I. Introduction

A. Context & Project Baseline

Industries and academic institutions are starting to rely more on hands-on project based teaching as it is recognized to be a better way to get familiar with the unknown. In the mechatronics industry, precise, low-cost experimental platforms for teaching control systems are becoming more in demand. This is a business opportunity that CedarMotion Technologies (CM Tech) must capitalize on as it represents a critical entry point for expansion into the European educational robotics sector. Having prior work with Vertical Take-off and Landing (VTOL) related controllers, CM Tech is at an advantage as the 1-DOF copter acts as a simplified pitch control subsystem. The platform is a thrust-actuated pendulum used commonly in academia to study non-linear dynamics and fundamental control problems. The 1-DOF copter will allow students to experience first-hand system modelling and identification similar to the process implemented in the industry in addition to designing and analysing controllers, such as a proportional-integral-derivative (PID) controller, based on specific requirements. Stabilizing the arm at a desired angle will inherently be a challenge due to the non-linear characteristics of the system, hence having an accurate model of the system is essential to design a position stability controller. To deploy the platform in European laboratories, standards and client specified technical requirements must be met to ensure a consistent platform under disturbance. The system motor must operate between 10-13 V_{system} and withstand temperatures between -10°C to 65°C all while remaining at a cost below 500$ and ensuring robustness and compliance with CE standards. Additionally, the project operates under a strict constraint of a 10-week delivery timeline, requiring rapid prototyping and validation. These requirements necessitate a mathematically precise representation of the system dynamics to ensure predictable and robust controller performance. The controller must meet specific measurable control performance requirements defined by the client.
Specifically the system must:

- Track a commanded angle within ±3° of the reference
- Achieve a settling time of no more than 1.5 seconds using the 5% criterion
- Limit the maximum overshoot to 15%

These qualifications represent the minimum acceptable performance for stable and predictable operation of the 1-DOF copter under varying conditions. They also establish the design targets that guide the modelling, analysis, and controller development presented in the subsequent sections of this report.

B. Literature Review

1) Fundamentals: The 1-DOF copter is widely used as a benchmark system in control education and research as its mechanically simple but dynamically rich thanks to its non-linear nature [1] , [2] . A typical aero-pendulum plant consists of a lightweight arm having a propeller supplied by a DC coreless motor at one end and fixed at the other. The angle is measured by a potentiometer at the pivot with the vertical axis as its reference [2] . The nonlinearity of the system originates from projecting the system's gravitational torque term on the vertical axis, which includes $sin(\theta)$. Moreover, thrust depends on motor voltage which is not linear due to the inductor found in the motor. Lastly even though the arm is connected through bearings, friction is still very present and making the system more sensitive to parameters [1] , [3] . The application of classic controllers require a linear system, thus most studies simplify the system by linearizing the system around small controllers, that is approximating the $sin(\theta) \approx \theta$ for small angles. Derived from taking the first order of the Taylor's expansion series.

$$sin(\theta) \approx \theta + \frac{\theta^3}{3!} + ... + \frac{\theta^{(2n+1)}}{(2n+1)!}$$

A transfer function is then derived by the linearization of Newton's laws [1] , [2] . After understanding the mechanics governing the plant, the derived TF is then used to compute the controller parameters: Kp, Ki, and Kd, for a PID controller

which is the most implemented type of controller. However most of the time the latter resulted in a poor transient response with high overshoot and relied on tuning techniques to achieve specified settling time. For example, Mohammadbagheri and Yaghoobi report settling times on the order of several seconds and high sensitivity to the selected PID gains [1] . These results are due to the linearization of a non-linear system with the assumption of initially having an accurate model of the plant. As the pendulum moves farther away from the equilibrium point the performance of the controllers degrade as the system becomes more sensitive to derived model parameters mismatch. Therefore a typical PID design will not be enough to stabilize a low cost plant while having stringent performance requirements. Hence more careful modelling a system identification are necessary. These observations motivate the use of more accurate modelling and systematic identification strategies, as discussed in the following subsection.

2) Modeling: Reports consistently stress on having an accurate mathematical model in order to design reliable controllers for aero pendulum. As mentioned before, due to the system's non-linear properties and sensitivity to parameter variations, the model must capture the relevant mechanics [2] , [3] . Most studies derive the dynamic equations using Newtonian mechanics, expressing torque balance about the pivot point (including gravity, thrust, and inertia) [3] . Small angle approximation is the most used method to linearize the system. Simply by replacing the $sin(\theta)$ term by θ the system becomes a linear one [1] , [2] , [3] . Having the linear dynamics equation, a transfer function (TF) related input motor voltage to desired angle position is derived [3] . The accuracy of the derived "white-box" model depends heavily on the correct estimation of physical parameters, hence small variations in these constants could significantly affect the predicted dynamics [4] . Several studies report discrepancies between simulated and experimental responses due to inaccurate parameter values or ignored effects such as pivot friction or aerodynamic drag [2] , [4] . System identification techniques, such as step and frequency response (FR) fitting, using experimental data, is the core to minimal model-plant mismatch [4] . The most effective strategy mentioned is the "gray-box" model, where the form of the TF is derived from the physics equations (white-box) but the parameter values are obtained using system identification (black-box) [4] .

3) Challenges & Justification: To address the system's non-linearities, several recent studies applied advanced optimization-based control strategies to overcome the limitations, operating beyond the linear region, of the classical PID controller [5] , [6] . Al-Ashtari present that particle swarm optimization (PSO) can automatically search for PID gains that minimize overshoot, settling time, and steady state error [5] . Their results show drastically improved transient performance compared to the derived gains for the plant. However, optimization based controllers require extensive computational resources, repeated simulations, and accurate cost functions, which can be impractical for low-cost educational platforms. Furthermore, Envikov introduces the H infinity controller for optimal robust performance ($H\infty$) that models the system using weighted functions to shape the closed loop (CL) behaviour across the low, medium, and high frequency ranges [2] . Given the project's constraints, high-complexity strategies are not directly suitable for deployment. Instead, reports suggest that accurate dynamics modelling with system identification provides a solid foundation for designing simpler controllers that still meet performance requirements. This motivates the controller design strategy adopted in this project, which relies on a validated model rather than purely trail-and-error gain tuning.

C. Report Outline

This report is organized as follows. Section II presents the white-box dynamic model of the 1-DOF copter, including the derivation of the governing equations, parameter estimation, and the linearized transfer function. Section III describes the experimental methodology and system identification procedures used to obtain an accurate plant representation and analyses the Open Loop (OL) response. Section IV describes the design process of the P, PID, and Lag-Lead controllers. Section V assesses and analyses the controller design alternatives based on the specified performance requirements. Section VI presents the implementation results and evaluates the closed-loop response. Finally, Section VII provides the main conclusions and outlines potential directions for future work.

II. Dynamic Model

In this section the derivation of the physics-based mechanical model of the 1-DOF copter based on Newton's second law for rotation about the pivot is done.

A. Equations of Motion & Parameter Definition

1) Plant Model: Unlike the plants discussed earlier in the literature review, this platform contains an additional counterweight on one end, increasing the system's sensitivity for angle above the reference horizontal axis.

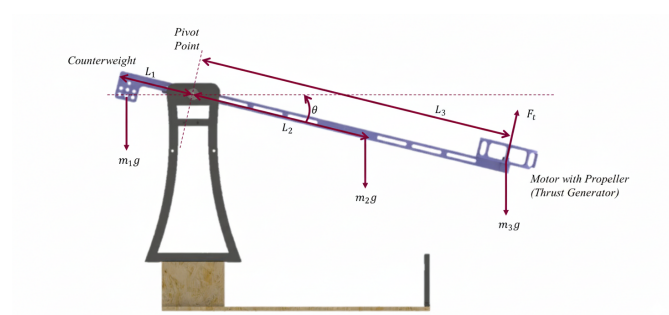

Fig. 1. Free-body diagram showing masses m_1, m_2, m_3, lever arms L_1, L_2, L_3, and thrust F_t about the pivot.

Torques taken into consideration acting on the system are gravitational torque, friction torque, and thrust torque from the motor. Applying the torque balance with the horizontal axis with CCW orientation as reference yields:

$$\sum \tau = J\ddot{\theta}$$

$$J\ddot{\theta}(t) + b\dot{\theta}(t) = g\cos\theta(t)\,(m_1L_1 - m_2L_2 - m_3L_3) + K_t\, i_m(t). \tag{1}$$

where:

- J: total moment of inertia about the pivot (kg m^2)
- b: viscous friction coefficient (N m s)
- $\theta, \dot{\theta}, \ddot{\theta}$: angular position, velocity, and acceleration (rad, rad/s, rad/s^2)
- g: gravitational acceleration (9.81 m/s^2)
- K_t: motor torque constant (N m/A)
- i_m: motor current (A)

The total inertia term is obtained by summing the inertias of the arm, motor, and counter weight:

$$J = \frac{m_2(L_1 + L_3)^2}{12} + m_1L_1^2 + m_2L_2^2 + m_3L_3^2 \tag{2}$$

Each component contributes based on its mass distribution and distance from the pivot. The non-linear nature of the system is clearly captured in Eq.1 through the $cos(\theta)$ term as a result of projecting the forces on the horizontal axis.

2) Motor Model: In addition to the plant dynamics, the circuit of the DC motor powering the propeller must be included in the model.

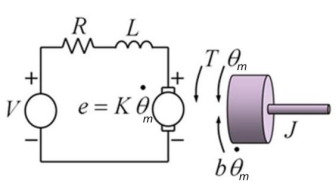

Fig. 2. DC motor circuit

$$v_m(t) = R_m i_m(t) + L_m \frac{d}{dt} i_m(t) + K_e\ddot{\theta}(t)$$

where:

- v_m: applied motor voltage (V)
- R_m: armature resistance (Ω)
- L_m: armature inductance (H)
- K_e: back-EMF constant (V s/rad)

with $V_{emf}(t) = K_e\ddot{\theta}(t)$, and $v(t) = v_m(t) - v_{emf}(t)$, hence:

$$v(t) = R_m i_m(t) + L_m \frac{d}{dt} i_m(t) \tag{3}$$

This model is essential to include since propeller thrust is directly proportional to the current ($F_{thrust} = k_t i_m$) which itself is related to the voltage supplying the motor.

B. Transfer Function Derivation

To derive a linear TF for the plant, the system is linearized around the horizontal hovering position ($\theta = 0$). The governing equation of motion, derived from Newton's Second Law, includes the inertial, damping, and gravitational moments:

$$J\ddot{\theta}(t) + b\dot{\theta}(t) + M_g\cos(\theta(t)) = k_t i_m(t)$$

where M_g represents the net gravitational moment coefficient derived from the mass imbalance ($M_g = g(m_1L_1 - m_2L_2 - m_3L_3)$). Around the horizontal equilibrium ($\theta \approx 0$), the small-angle approximation $\cos(\theta) \approx 1$ is applied. This results in a constant gravitational torque term that acts as a static disturbance:

$$J\ddot{\theta}(t) + b\dot{\theta}(t) + M_g = k_t i_m(t)$$

This static gravitational constant is compensated by applying a constant voltage bias (V_{bias}). The total motor voltage is defined as the sum of this bias and the control effort $v(t)$. So it will be ignored in the TF and be later added in the simulation circuit as a bias. The bias voltage is selected specifically to counteract the static gravity at equilibrium ($\ddot{\theta} = \dot{\theta} = 0$).

$$J\ddot{\theta}(t) + b\dot{\theta}(t) = k_t i_m(t)$$

This implies that the dynamic relationship between the control voltage $v(t)$ and the angular position $\theta(t)$ is not affected by the constant gravity term. Taking the Laplace transform (assuming zero initial conditions) yields:

$$(Js^2 + bs)\Theta(s) = k_t I_m(s) \tag{4}$$

Applying the Laplace Transform of the motor model (Eq. 3)

$$V(s) = (R_m + L_m s)I_m(s)$$

However, The inductance of a coreless DC motor is extremely small (typically on the order of 10^{-6} H). Its effect on the armature current dynamics is negligible because the electrical time constant ($\tau_e = L_m/R_m$) is orders of magnitude faster than the mechanical time constant. Consequently, the term $L_m s$ is omitted without altering the dominant system response. The simplified current equation becomes:

$$I_m(s) = \frac{V(s)}{R_m} \tag{5}$$

Substituting the electrical relationship (Eq. 5) into the Laplace transform of the mechanical equation (Eq. 4):

$$Js^2\Theta(s) + bs\Theta(s) = K_t\left[\frac{V(s)}{R_m}\right] \tag{6}$$

The final theoretical Transfer Function relating input voltage to output angle is:

$$G(s) = \frac{\Theta(s)}{V(s)} = \frac{\frac{K_t}{JR_m}}{s\left(s + \frac{b}{J}\right)} \tag{7}$$

where $K = \frac{K_t}{JR_m}$. The resulting theoretical model is a Type 1 system (containing a pure integrator). This indicates that mathematically, a constant input voltage change results in a constant angular velocity. This derivation serves as the baseline white-box model, with the discrepancy addressed by adding the voltage bias.

C. Parameter Estimation

This section details the experimental "gray-box" process used to find the numerical values of the unknown parameters in the theoretical model and to develop a final, high-fidelity transfer function of the plant.

1) Total Moment of Inertia (J): It is the simple sum of the moments of inertia of each individual part about a specific axis (the pivot point)

$$J_{total} = J_{assembly} + J_{counterweight} + J_{motor}$$

$$J_{total} \approx 0.005135(kg.m^2)$$

2) Torque Constant (K_t): The torque constant is determined using the static torque balance equation at the horizontal hover point(θ= 0). The required thrust torque to counteract the known gravitational torque is calculated using the physical parameters from the project appendices. By experimentally finding the current required to hold the arm at this position ($i_{m,hover}$), K_t is calculated as $K_t = \frac{\tau req}{i_{m,hover}}$, yielding a value of $K_t \approx 0.0183$N.m/A

3) Friction Coefficient (b): From different experiments and reports, for a pendulum with bearings it is safe to assume a friction coefficient of 0.01 N m s/rad

4) Armature Resistance (Rm): The motor's internal resistance is found by applying a range of steady-state voltages and measuring the corresponding currents while the arm is hovering. At steady state the inductor acts as a short circuit and the back emf is zero, since its related to the angular velocity when the arm is hovering on a fixed point. A linear regression on the V-I data is performed to find the slope ($R_m = \frac{\Delta V}{\Delta I}$), which isolates the resistive component. The analysis of 16 data points yielded a high-confidence value of $R_m = 0.8818\Omega$.

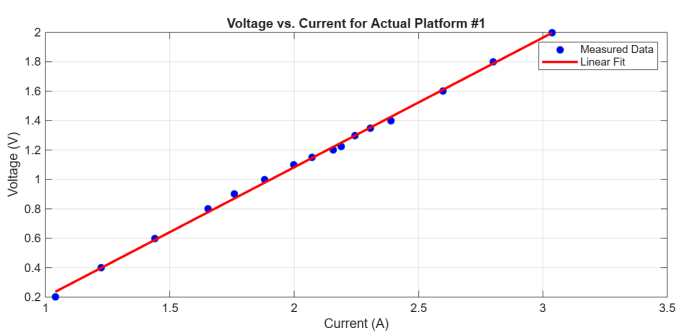

Fig. 3. Example V–I curve used to estimate R_m.

5) Armature Inductance (Lm): Based on the previous assumption of Eq.10, $L_m = 0$.
Values of Parameters:

- J_{total}: 0.005135 (kg m^2)
- K_t: 0.0183 (N m/A)
- b: 0.01 (N m s/rad)
- R_m: 0.8818 (Ω)
- L_m: 0 (H)

D. Final Plant Model

Filling the numerical values into the TF (Eq. 7):

$$G_{white-box} = \frac{4.0414}{s\,(s + 0.194)} \tag{8}$$

with a poles located at $p_1 = 0$ (pure integrator) and $p_2 = -0.194$ with an infinite system gain (Type 1 system). This second-order approximation captures the dominant behaviour of the plant and serves as the reference model for the following analysis. Although the model is analytically derived rather than identification-based, the next section verifies that its predicted order and qualitative response are consistent with the experimentally observed dynamics.

III. System Analysis

This section analyses the behaviour of the 1-DOF plant. From supplying a known input into the unidentified "black-box" system and returning an output, enabling the derivation of the TF through MATLAB's System Identification toolbox. The tool box performs curve fitting on a pre-specified data set depending on the number of zeros and poles given. Since the system's white-box model already is derived, the TF must ultimately be in the form of a 2nd Order Type 0 system; that is 2 poles with no zeros.

A. Time Domain (Step) Response

The first approach is applying a voltage bias (at t=0) to stabilize the arm around 0° (horizontal axis taken as reference CCW sense). After reaching steady state (at t =10), a positive step input is added increasing the arm's angle. The data set is measured from t=0 to t=20 using a 0.01 sampling rate. After fitting the data set to a 2nd order TF with no zeros getting the following TF with a 95% fit:
Resulting in the following TF:

$$G(s)_{step} = \frac{8.209}{s^2 + 4.642s + 16.83}$$

Applying the final value theorem (FVT) gives us a System gain of 0.483 and pole location of $p_{1,2} = -2.32 \pm j3.38$

B. Frequency Response

Similar to the step response, a bias voltage is added. Some reports suggests using the "chirp" block in Simulink which automatically changes the input sine wave frequency over a determined time interval. This method does not allow the system to reach steady state before moving to a higher frequency. Thus manual frequency sweep is done over a set of low, medium,

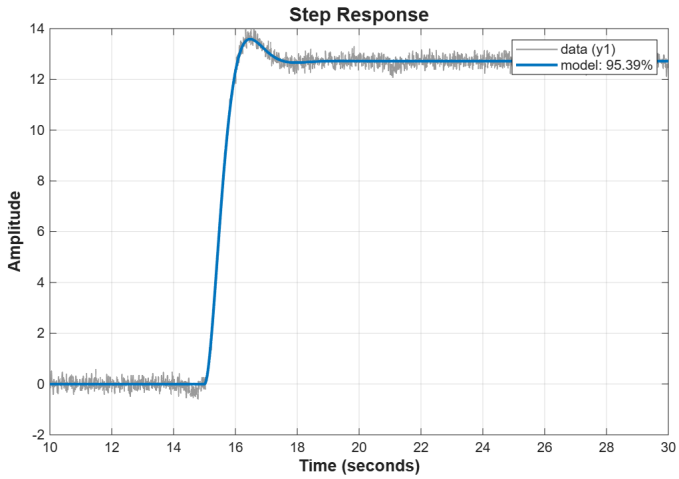

Fig. 4. Step response graph used to identify the "black-box" model.

and high frequency inputs to accurately model the system's behaviour over all applications. The system is subjected to the voltage bias in addition to a sine wave of 0.3 amplitude of different frequencies. At each frequency, measurements of the average of peak-to-peak magnitude is taken after steady state is achieved. Using this data to derive a bode plot with 70% fit estimate which in turn gives us the transfer function of the system.

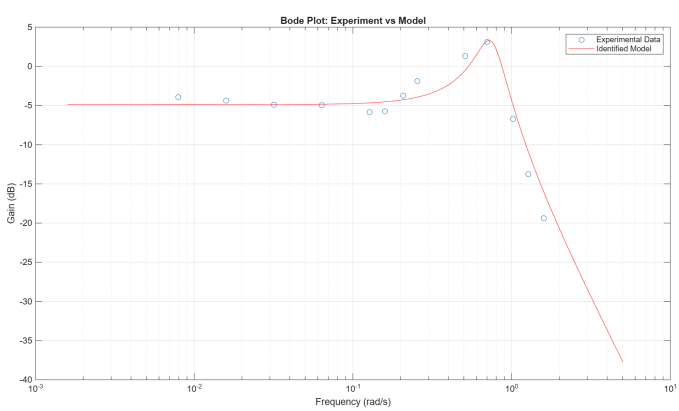

Fig. 5. Bode Plot using the measured data of the frequency response.

Locating a slope of $-40dB/decade$ that shows a pair of complex conjugate poles with a System gain of 0.569 and poles located at $p_{1,2} = -0.9 \pm j2.7$. Thus the TF:

$$G(s)_{\text{frequency}} = \frac{12.68}{s^2 + 1.873s + 22.25} \tag{9}$$

C. Delay Estimation

A critical parameter for accurate modelling and control is the system's "dead time" or system delay. This is the time between a command being issued and the physical system beginning to react. This delay arises from a combination of computational processing, USB communication, and actuator lag. Furthermore, the Low Pass Filter (LPF) applied to the angle sensor output contributes significantly to this total delay. While the filter is necessary to attenuate measurement noise fed-back into the controller, it inherently introduces a phase lag that increases the effective system dead time. To measure this delay, a small-perturbation step test is conducted. A step voltage is applied at a known time (t = 10s), and the system's response is recorded with a high sampling rate. By plotting the input voltage and the output angle on the same time axis and zooming in on the moment the step is applied, the delay is measured as shown in Fig. 6.

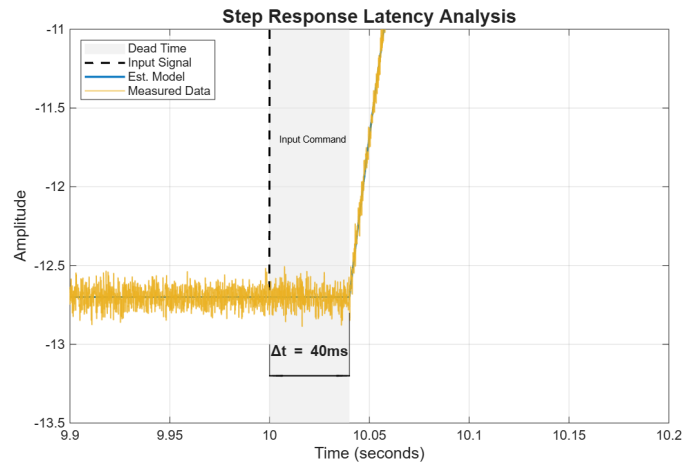

Fig. 6. Experimental estimation of the system delay.

The time interval between the step instant (t = 10 s) and the moment when the arm angle first starts to deviate from its initial value is approximately 40 ms. Therefore model the plant delay as T_d = 0.04 s. In the Laplace domain, this delay is represented by the factor $e^{-T_d s}$. To incorporate this term into our linear transfer–function model, using the first–order Padé approximation:

$$e^{-T_d s} \approx \frac{1 - \frac{T_d}{2}s}{1 + \frac{T_d}{2}s} = \frac{1 - 0.02s}{1 + 0.02s}.$$

This approximation introduces a right–half–plane zero at

$$s_z = \frac{1}{0.02} = +50 \text{ rad/s},$$

and a left–half–plane pole at

$$s_p = -\frac{1}{0.02} = -50 \text{ rad/s}.$$

These pole–zero locations are used in the final plant model employed for controller design.

D. System Gain Correction

To ensure the robustness of the controller implementation and address potential discrepancies between the theoretical model and the physical hardware, an experimental verification of the system's system gain is conducted. A step response test is performed around the hovering equilibrium point ($V_{bias} \approx$ 1.1V) by applying a small voltage perturbation and measuring the resulting steady-state angular displacement. This informative study revealed a physical system gain of $K_{real} \approx 0.71$, which

is significantly higher than the initial model's gain of $K_{inital} \approx 0.57$ (from Eq. 9). This indicates that the actual actuator is approximately 1.4 times more sensitive than initially modelled.

$$\alpha = \frac{K_{model}}{K_{real}} = 0.7125$$

To account for this, the plant transfer function is updated by scaling the numerator to match empirical reality:

$$\text{System Gain}_{\text{new}} = \frac{12.68}{\alpha} \approx 17.8$$

The final identified model, including the Padé delay and updated system gain , is therefore:

$$G_{\text{grey-box}}(s) = \frac{17.8(50 - s)}{(s + 50)(s^2 + 1.873s + 22.25)} \tag{10}$$

E. Open-Loop Analysis

Even though the fit estimate of the step response is higher than that of the frequency response, the latter approach provides higher fidelity as it excites the system across a spectrum of frequencies, thus describing the system's dynamics more accurately. Therefore, the controller design is built upon the transfer function derived from the frequency response. The open-loop properties of the identified plant, including the 40 ms time delay approximated by a first-order Padé function, are analysed to establish a performance baseline and quantify the control requirements.

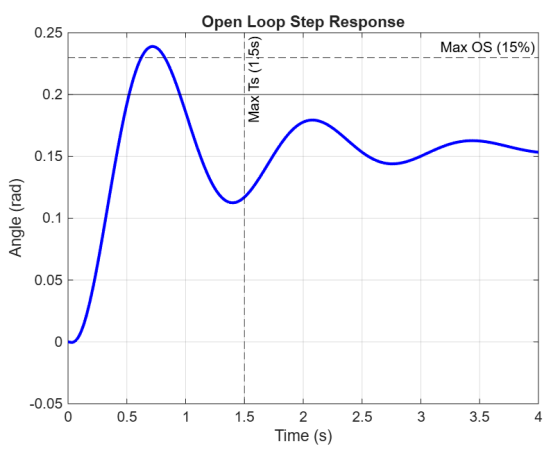

Fig. 7. Open-loop step response showing Type 0 behaviour and finite steady-state error.

1) Time Domain Behaviour: The open-loop step response is presented in Fig. 7. The system exhibits an underdamped response with a finite steady-state value. This behaviour highlights a critical divergence from the ideal derivation in Section II, which predicted a Type 1 system (pure integrator). The experimental response settles to a constant angle below the desired angle for the given voltage, identifying the physical plant as a Type 0 system. This confirms that the real-world setup is not perfectly modelled, due to the model-plant discrepancies and approximations done. This results in a finite system gain of 0.8, 50% overshoot (w.r.t steady-state angle), a settling time around 4 seconds, and the inability to track a step reference without error.

2) Frequency Domain Analysis: The frequency response characteristics are evaluated using the Bode diagram shown in Fig. 8 and summarized in Table I.

a) Phase Margin (PM): The system possesses a Phase Margin of 26.4° at a crossover frequency of 5.96 rad/s. This low phase margin indicates a poor damping ratio ($\zeta \approx \frac{PM}{100} \approx 0.26$), which directly explains the large oscillatory overshoot observed in the time domain. To meet the overshoot specification (< 15%), the Phase Margin must be improved to approximately 60°.

b) Gain Margin (GM): The Gain Margin is calculated as 8.85 dB. While the system is stable, this falls below the typical robust design target of 10 dB, indicating limited robustness against changes in system gain or actuator performance.

c) Bandwidth & Delay: The open-loop bandwidth is 7.12 rad/s. However, the phase curve exhibits a sharp roll-off at higher frequencies. This is attributed to the 40 ms time delay, which adds a phase lag proportional to frequency ($\phi_{delay} = -\omega T_d$). This delay imposes a fundamental limit on the achievable closed-loop bandwidth.

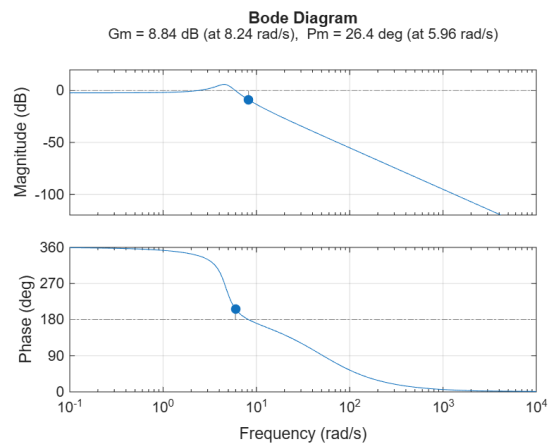

Fig. 8. Open Loop Response Bode Plot

TABLE I
OPEN-LOOP PERFORMANCE SUMMARY

Metric	Value	Requirement	Status
System Type	Type 0	Type 1 (Ideal)	Mismatched
Phase Margin	26.4°	≈ 60° (Target)	Poor
Gain Margin	8.85 dB	> 10 dB	Marginal
Bandwidth	7.12 rad/s	N/A	Suffecient

IV. CONTROL SYSTEM DESIGN

To guarantee predictable transient behaviour, the time-domain performance specifications are translated into equivalent target metrics for both the s-plane (Root Locus) and the frequency domain (Bode). The project constraints require a maximum overshoot (Mp) of less than 15% and a settling time (Ts) of under 1.5 seconds.

For the Root Locus-based design, these constraints are mapped to a conservative damping ratio of $\zeta = 0.6$ and a natural

frequency of $\omega_n = 5.0$ rad/s, establishing the desired dominant closed-loop pole location at $s_{desired} = -3 \pm j4$. Complementary to this, for the FR design, these damping and speed requirements are translated into a target Phase Margin ϕ_m of approximately 60° (based on the approximation ($\zeta \approx \phi_m/100$) and a closed-loop bandwidth (ω_{BW}) exceeding 6 rad/s. Additionally, a Gain Margin (Gm) of at least 10 dB is specified to ensure robustness against parameter uncertainty. These dual-domain targets serve as the primary design objectives for the PID and Lag-Lead controllers developed in the subsequent subsections.

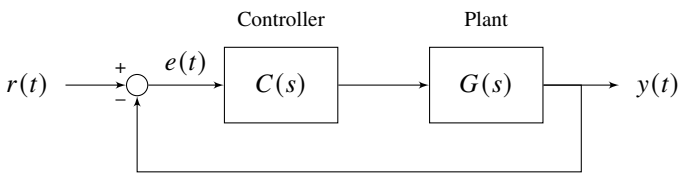

Fig. 9. Unity-feedback block diagram with controller $C(s)$ and plant $G(s)$.

First the design and analysis of a simple proportional (P) controller is done. This establishes a performance baseline and justifies the need for a more complex compensator. A proportional controller,$C_P(s) = Kp$, is the simplest form of feedback. The closed-loop TF is given by T(s):

$$T(s) = \frac{K_p G(s)}{1 + K_p G(s)}$$

Where G(s) is the final 3rd-order plant model with the Padé approximation for the time delayed and improved system gain (Eq. 18). The system's behaviour is determined by the root of the closed loop characteristic equation $1 + K_p G(s) = 0$.

1) Stability Analysis: Values of Kp for which the closed loop remains stable must be identified. The Nyquist Criterion (N = Z + P), requires the number of encirclements of the critical point (-1, j0) to be equivalent to the number of OL poles in the RHP (P) subtracted from the number of unstable CL poles (Z). The maximum stable gain can be computed from evaluating the reciprocal of the gain of G(s) at the cut-off frequency.

$$K_{max} = \frac{1}{|G(jw_c)|} = \frac{1}{0.370}$$
$$\therefore K_p \leq 2.702$$

Fig.10 illustrates this result using the Nyquist plot generated in MATLAB.

2) Steady-State Error Analysis: The identified plant is a Type 0 system, which will have a finite steady-state error (e_{ss}) for a step input. The formula for this error is obtained by applying FVT to the error signal.

$$e_{ss} = \lim_{s \longrightarrow 0} E(s) = s\frac{R/s}{1 + K_p G(s)}$$

Where R is the magnitude of the step input. The system gain of the plant is G(0) = 0.8. The project requires the final angle to be within ±3° of the target. For a typical step command of 15° , the error must be less than 3°.

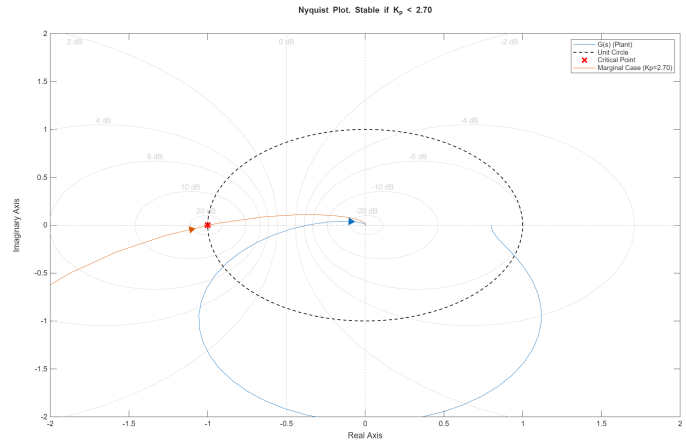

Fig. 10. Nyquist Plot showing the stability limit at $K_p = 2.7$

$$e_{ss} = \lim_{s \longrightarrow 0} E(s) = \frac{15°}{1 + K_p * 0.8} \leq 3°$$
$$K_p \geq 5$$

3) Contradiction: From both analyses discussed above the following conditions were derived:

$$K_p \geq 5$$
$$K_p \leq 2.702$$

No intersection between these two intervals exists, thus no values of K_p will be able to meet the specified requirements. To further validate that a P controller is not feasible, Fig 11 shows the Root Locus plot of G(s) illustrating that no value of Kp will make the root locus pass through the desired pole locations.

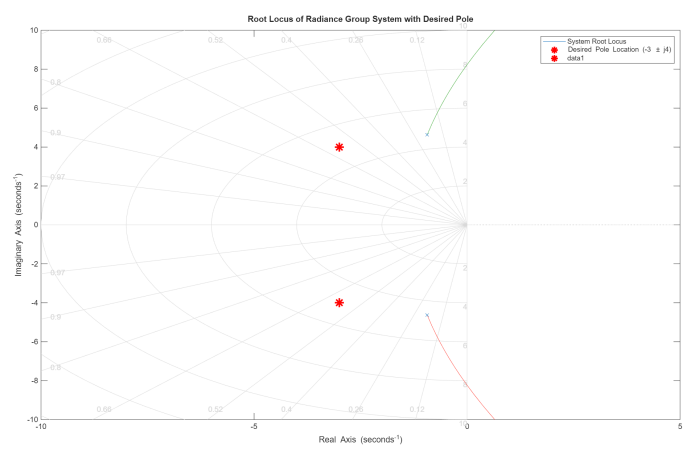

Fig. 11. Root Locus showing no possible value of K_p will achieve the desired requirements

4) Design for Stability: To establish a performance baseline, a Proportional (P) controller is designed with the primary objective of ensuring closed-loop stability. Root locus analysis indicates that the system damping ratio (ζ) is inversely proportional to the controller gain (K_p). Consequently, the

overshoot constraint ($\zeta = 0.6$) cannot be met with simple proportional action, as the open-loop damping is already below 0.2 (Fig. 9). Therefore, the design criterion is shifted to maximize stability margins. Based on the critical gain of $K_{max} \approx 2.7$ identified in the stability analysis, a gain margin of 6 dB is selected to ensure maximum stability. This yields a design gain of:

$$K_p = \frac{K_{max}}{2} = 1.35$$

The theoretical performance of this "stable" P-controller ($K_p = 1.35$) is evaluated against the project specifications:

1) Steady-State Error: With a system gain of 0.8, the predicted error for a 15 step is 7.2°, which violates the ±3° accuracy requirement.
2) Transient Response: The resulting damping ratio drops to $\zeta \approx 0.14$, predicting an excessive overshoot of approximately 65%.

While the P-controller with Kp =1.35 yields a stable closed-loop system, it fails to meet both the accuracy and transient response specifications. Increasing the gain to reduce error would drive the system to instability, while decreasing the gain to improve stability would further increase the steady-state error. This necessitates the implementation of a more complex controller to decouple the steady-state accuracy from the transient stability.

A. PID Controller Design

Achieving the required performance of the system is non-negotiable. Eliminating steady-state error and oscillatory response is critical. A PID controller is designed using the Root Locus and FR methods to allow independent control of damping and natural frequency.

$$C_{PID}(s) = K_c \frac{(s+z_1)(s+z_2)}{s}$$

1) Design via RL Method: First designing the PD component and afterwards the PI component with minimal effect on the conditions satisfied by the first step of design. The angle deficiency at the desired pole location s_d, found in the beginning of this section, is calculated by summing the angular contributions of the OL poles (plant complex poles).

$$\sum \angle\theta_{zeros} - \sum \angle\theta_{poles} = (2k+1)180°$$

Ignoring the angle contribution of the Padé pole and zero since they are very far with respect to the point of interest (x10 farther from s_d along the x-axis) and their angle deficiency almost adds up to 180° The zero location of the PD component must contribute to satisfy the angle condition.

$$\angle\theta_{pd} = 76.4°$$

Hence z_{pd} = -3.97.
To introduce the integral action without distorting the root locus branches, the PI zeroz_{pi} is placed conservatively near the origin at s=-0.2.
The required OL gain K is determined by applying the magnitude condition at s_d, ensuring the root locus passes exactly through the desired points.

$$K = \left| \frac{(s+50)(s^2+1.873s+22.25)}{17.8(50-s)} \right|_{s_d=-3+j4}$$

$$K = 4.2$$

$$K_c = \frac{4.2}{17.8} = 0.235$$

Now the PID parameters can be computed:

$$C_1(s) = 0.235 \frac{(s+3.97)(s+0.2)}{s} \tag{11}$$

Resulting in the following PID Parameters:

- $K_p = 0.979$
- $K_d = 0.235$
- $K_i = 0.182$

2) Design via Frequency Response: Another method of designing a PID-Controller is through the FR method. Both designs must yield to PID parameters that achieve the time-domain requirements. The design process is consists of an initial conservative design based on phase margin (PM) requirements. The performance requirements are translated into frequency domain targets. The overshoot constraint (< 15%) corresponds to a damping ratio $\zeta \approx 0.52$, which maps to a required Phase Margin (Φ_m) of approximately 55° to 60°. The settling time constraint ($T_s < 1.5$ s) implies a minimum closed-loop bandwidth. Assuming a standard second-order approximation, the natural frequency ω_n must satisfy $\zeta\omega_n \geq 2$, suggesting a minimum target crossover frequency $\omega_{gc} \approx 4$ rad/s. The controller structure is selected as a "double-zero" PID to provide maximum phase lead:

$$C(s) = K \frac{(s+z)^2}{s}$$

where the term $1/s$ provides integral action to eliminate steady-state error, the term $(s+z)^2$ provides phase lead, and K sets the bandwidth. For the initial design, the target crossover frequency is set to $\omega_{gc} = 4$ rad/s. The FR of the plant combined with the integrator ($G(s)/s$) is evaluated at this frequency. The uncompensated phase is found to be approximately −149.4°. The desired PM of 60° at the crossover frequency requires a −120° OL phase. Hence to achieve a target PM, the controller zeros must provide a phase boost of approximately +30° (that is 15° compensated per zero). Solving for the zero location z that provides this phase contribution at $\omega = 4$ rad/s yields $z = 15$. The required gain K is then computed to enforce unity loop gain ($|C(j4)G(j4)| = 1$), resulting in $K \approx 0.0091$.

$$C_2(s) = 0.0091 \frac{(s+15)^2}{s} \tag{12}$$

Resulting in the following PID Parameters:

- $K_p = 0.273$
- $K_d = 0.0091$
- $K_i = 2.05$

B. Lag-Lead Compensator Design

As a design alternative, a Lag-Lead compensator is also implemented with the same transient and steady-state requirements.

1) Design via RL Method: Similar to the PID RL design, the RL will be forced to pass by the desired pole location s_d through satisfying the angle and magnitude same conditions. The total angle contribution of the OL plant, including the Padé pole and zero, at s_d is calculated to be $-309.5°$. hence to satisfy the angle criterion $\angle G(s)H(s) = -180°$, a total phase lead of 129.5° is required. It is usually recommended to limit a lead network's compensation to 70° to avoid having far poles and zeros in the RL. Hence, a double lead structure,two identical cascaded lead stages, will be implemented, where each network compensates for half the deficiency. ($\phi_{each} \approx 64.75°$) To maximize efficiency, the lead zero is placed directly below the real part of the desired pole at $z_{lead} = -3$. The corresponding pole location is derived to geometrically satisfy the angle contribution.

$$\angle(s_d + z_{lead}) - \angle(s_d + p_{lead}) = 64.75°$$

Solving this yields a pole location at $p_{lead} \approx -11.58$. this contributes to a total lead of 138.9°, providing a small safety margin over the required 129.5° .

The controller gain K_c is computed using the magnitude condition at s_d.

$$|G_{plant}(s_d)| \cdot |G_{lead}(s_d)|^2 \cdot K = 1$$

Evaluating the magnitude yields a required gain $K \approx 5.32$. to satisfy the steady-state error specification, the system gain must be boosted. the current compensated system gain is around 0.286 whereas the requirements dictates a gain of roughly 20. this necessitates a gain boost factor of $\alpha \approx 70$.

to implement this without having an effect on the transient root locus design, the lag pole and zero are placed very close to the origin.

$$p_{lag} = -0.001, \quad z_{lag} = \alpha \cdot p_{lag} = -0.07$$

the angle contribution of this pair log at s_d is negligible $-0.6°$, preserving the stability margins established by the lead network.

The complete analytical transfer function for the Lag-Lead compensator using the RL method is:

$$C_3(s) = 5.32 \left(\frac{s+3}{s+11.58}\right)^2 \left(\frac{s+0.07}{s+0.001}\right) \tag{13}$$

2) Design via Frequency Response: A Lag-Lead compensator is also designed but using FR techniques. While the method utilized s-plane geometry to position closed-loop poles, this approach focuses directly on shaping the open-loop Bode magnitude and phase characteristics to satisfy the stability margins and bandwidth requirements.

To satisfy the settling time specification ($T_s < 1.5$ s), the theoretical minimum bandwidth for a second-order system is calculated as $\omega_{n,min} \approx 4/(0.6 \times 1.5) \approx 4.5$ rad/s. However, designing strictly at this theoretical limit is deemed insufficient for two reasons:

1) Transport Delay: The system includes a 40 ms delay, which reduces the effective phase margin.
2) Lag Compensation Effects: The addition of a lag compensator (required for steady-state accuracy) introduces a pole-zero pair near the origin, which is analytically known to create a slow settling "tail" in the time response if the lag zero is placed at too low frequencies.

To robustly guarantee compliance with the settling time constraint despite these unmodeled dynamics, a design safety factor of 3× is applied to the bandwidth target. Consequently, the gain crossover frequency is selected as:

$$\omega_c \approx 3 \times \omega_{n,min} \approx 14.5\,\text{rad/s}$$

This target frequency (ω_c) is used as the basis for calculating the lead and lag parameters. Additionally, to satisfy the overshoot constraint (< 15%), a target Phase Margin of approximately 60° is required. The plant model $G(s)$ (including the Padé delay) is evaluated at the target crossover frequency:

$$|G(j14.5)| \approx -20.6\,\text{dB}, \quad \angle G(j14.5) \approx -204°$$

To achieve a Phase Margin of 60°, the target phase at crossover must be $-180° + 60° = -120°$. Consequently, the required additional phase is calculated as:

$$\Phi_{needed} = -120° - (-204°) = +84°$$

A single lead stage is practically limited to providing approximately 60° of phase advance. Since the required boost is 84°, a Double Lead structure is employed. To maximize phase efficiency and lift the phase dip prior to crossover, the lead zero is placed at $z_{lead} = 4$ rad/s. The corresponding pole location is selected to limit high-frequency noise amplification while ensuring sufficient phase lead, resulting in $p_{lead} = 38$ rad/s. This configuration provides robust phase support in the 4 to 40 rad/s range.

A lag network is required to boost the low-frequency gain for steady-state accuracy. In standard textbook designs, the lag zero is typically placed one decade below the crossover frequency (e.g., 1.45 rad/s). However, this often results in a slow closed-loop pole (time constant $\tau \approx 1/z_{lag}$), causing the settling time violation.

To eliminate this "slow tail," an aggressive lag zero placement strategy is adopted. The lag zero is placed at $z_{lag} = 6$ rad/s. While this is closer to the crossover frequency, the ample phase margin provided by the double-lead network accommodates the slight phase penalty, ensuring that the error dynamics settle rapidly ($T_{settle} \approx 1/6$ s). The lag pole is placed at $p_{lag} = 0.05$ to provide a high system gain ratio ($\alpha = 120$).

Finally, the scalar gain K is computed to force the magnitude of the total loop $L(s)$ to be unity (0 dB) at the target crossover frequency $\omega_c = 14.5$ rad/s:

$$K = \frac{1}{|G_{plant}(j14.5) \cdot G_{lead}(j14.5)^2 \cdot G_{lag}(j14.5)|} \approx 72$$

The final Lag-Lead compensator transfer function is:

$$C_4(s) = 72\left(\frac{s+4}{s+38}\right)^2\left(\frac{s+6}{s+0.05}\right) \tag{14}$$

C. Controller Design Summary

Table II summarizes the transfer functions for the four distinct control strategies developed in this section. These controllers represent the candidates for the simulation analysis and hardware implementation presented in the subsequent sections.

TABLE II
SUMMARY OF DESIGNED CONTROLLERS

Control Strategy	Controller Transfer Function
Proportional (P)	1.35
PID (RL)	$0.235\frac{(s+3.97)(s+0.2)}{s}$
PID (FR)	$0.0091\frac{(s+15)^2}{s}$
Lag-Lead (RL)	$5.32\left(\frac{s+3}{s+11.58}\right)^2\left(\frac{s+0.07}{s+0.001}\right)$
Lag-Lead (FR)	$72\left(\frac{s+4}{s+38}\right)^2\left(\frac{s+6}{s+0.05}\right)$

V. SIMULATION RESULTS

To validate the design methodologies and ensure compliance with the performance specifications ($T_s \leq 1.5$ s, $M_p \leq 15\%$, $e_{ss} = 0$), non-linear time-domain simulations are conducted. Unless otherwise specified, all step response simulations utilise a reference command amplitude of $R = 0.2$ radians ($\approx 11.5°$). This value represents a typical operating manoeuvre within the linear region of the actuator. Furthermore, the simulations incorporate the physical saturation limits of the actuator (+1.9 V and −1.1 V relative to the bias) and the anti-windup clamping logic to ensure fidelity to the physical platform.

A. P-Controller Assessment

The response of the robust P-controller ($K_p = 1.35$), designed for a 6 dB gain margin, is shown in Fig. 12. As predicted by the stability analysis, the system remains stable but exhibits a large steady-state error of approximately 7.2° and a sluggish rise time. This simulation confirms that proportional action alone is insufficient to decouple stability margins from accuracy requirements.

B. PID Controller: Root Locus Design

The evolution of the PID controller parameters derived via the Root Locus method is illustrated in Fig. 13. The design process involves four distinct iterations to resolve the conflict between stability and settling time. The progression of parameters and their effects on the system response are summarised in Table III.

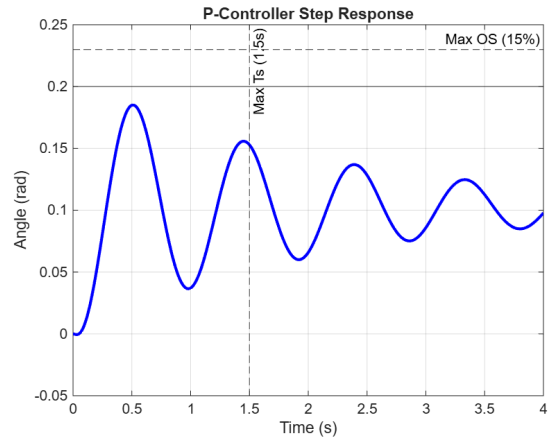

Fig. 12. P-Controller response ($K_p = 1.35$) demonstrating stability but failing accuracy requirements.

TABLE III
PID DESIGN ITERATIONS (RL METHOD)

Iter.	Configuration	Response Observation
1	Analytical Angle Match ($z_{PI} = -0.2$)	Stable, but exhibits a "long tail" due to the pole-zero dipole.
2	Dipole Elimination ($z_{PI} \to -1.6$)	The slow tail is eliminated, but the rise time remains too slow.
3	Bandwidth Expansion	Rise time improves, but overshoot exceeds 20% due to reduced damping.
4	Active Damping (K_d increased)	**Final Design.** Overshoot reduced to 10.5%.

The final iteration parameters ($K_p = 2.1, K_i = 3.5, K_d = 0.65$) yield a stable response with 10.48% overshoot. The settling time is 1.516 s, which is marginally above the strict 1.5 s.

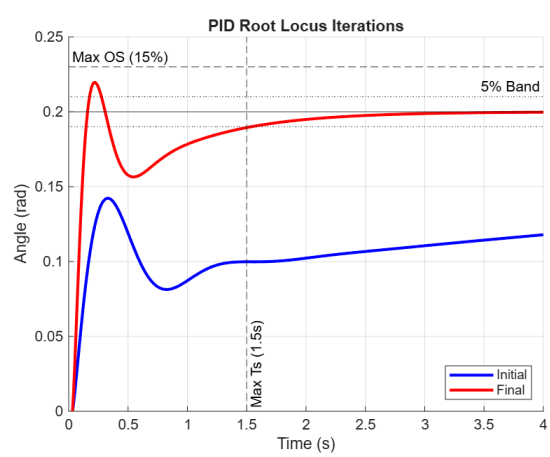

Fig. 13. Evolution of PID step response (Root Locus Design) from initial dipole-dominated response to final agile response.

C. PID Controller: Frequency Response Design

A parallel design effort is conducted using Frequency Response techniques (Bode shaping). Fig. 14 compares the initial and final responses. The iterative tuning process is detailed in Table IV.

TABLE IV
PID Design Iterations (FR Method)

Iter.	Strategy	Response Observation
1	Phase Margin Target (60°)	Stable but sluggish; bandwidth is insufficient for 1.5 s settling time.
2	Increase Crossover	Rise time improves, but Phase Margin drops.
3	Adjust K_d	**Final Design.** Specifications met with 1.500 s settling time.

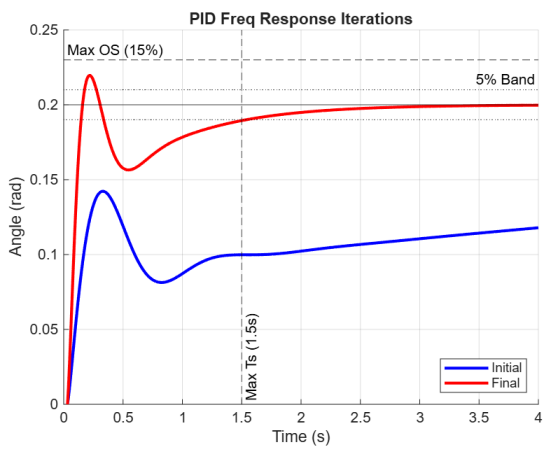

Fig. 14. Comparison of Initial and Final PID designs using Frequency Response methods.

D. Lag-Lead Compensator: Root Locus Design

The Lag-Lead compensator is initially designed using Root Locus geometry. Fig. 15 displays the step response improvement. As shown in Table V, the primary challenge is balancing the steady-state error correction (Lag) with the transient damping (Lead).

TABLE V
Lag-Lead Design Iterations (RL Method)

Iter.	Strategy	Response Observation
1	Conservative Lag Zero	High overshoot observed; Lag network interacts with dominant poles.
2	Shift Lead Zero Left	**Final Design.** Improved damping ratio. Overshoot reduced to $< 15\%$.

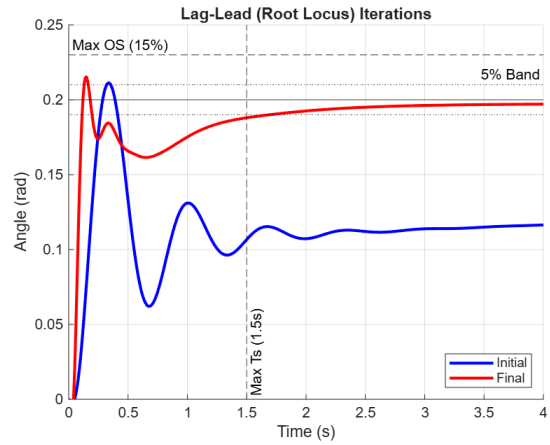

Fig. 15. Root Locus-based Lag-Lead design iterations.

E. Lag-Lead Compensator: Frequency Response Design

Fig. 16 presents the response of the analytically manipulated Lag-Lead compensator ($K = 72, z_{lag} = 6$). Unlike the previous designs, this controller satisfies all performance requirements without iteration. By proactively incorporating a 3× bandwidth safety factor and an aggressive lag zero placement, the "slow tail" effect is pre-empted. The system achieves a settling time of 1.439 s, which is the fastest compliant time among all designs.

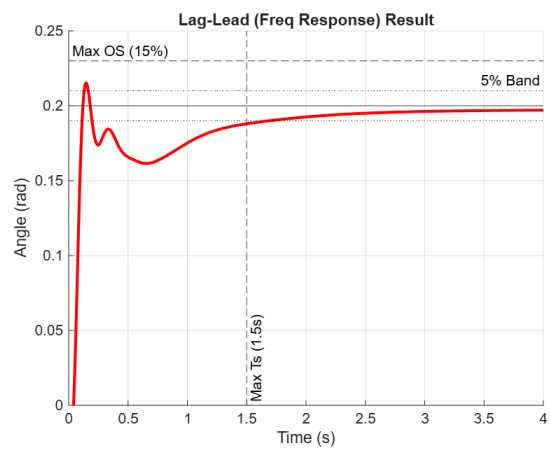

Fig. 16. Step response of the Direct Synthesis Lag-Lead compensator (Frequency Response Design).

F. Frequency Domain Analysis & Robustness

To verify the stability margins and dynamic capabilities of the designed controllers, the OLFR is analysed. The Bode plots for the three final designs are presented in Fig. 17, Fig. 18, and Fig. 19.

1) Stability Margins: The robustness of the control system is quantified by the Gain Margin (GM) and Phase Margin (PM).

- PID (Root Locus): This design (Fig. 17) achieves a Phase Margin of 48.25° and a Gain Margin of 13.12 dB.
- PID (Freq. Response): This design (Fig. 18) achieves a PM of 44.98°, which is the lowest among the three, correlating with the highest overshoot (14.85%).

- Lag-Lead: The Lag-Lead design (Fig. 19) exhibits a superior Phase Margin of 61.03° at 14.47 rad/s, confirming the robustness of the frequency-domain design.

2) Bandwidth: The closed-loop bandwidth (ω_{BW}) serves as a proxy for response speed. The Lag-Lead compensator achieves a significantly higher bandwidth (41.58 rad/s) compared to the PID controllers (≈ 23 rad/s). This extended bandwidth explains how the Lag-Lead design achieves the fastest settling time while maintaining the highest stability margins.

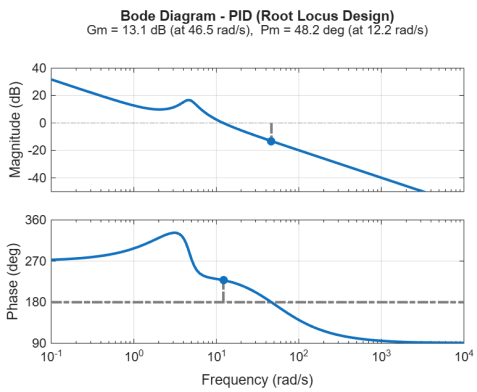

Fig. 17. Open-Loop Bode Diagram for PID (Root Locus Design).

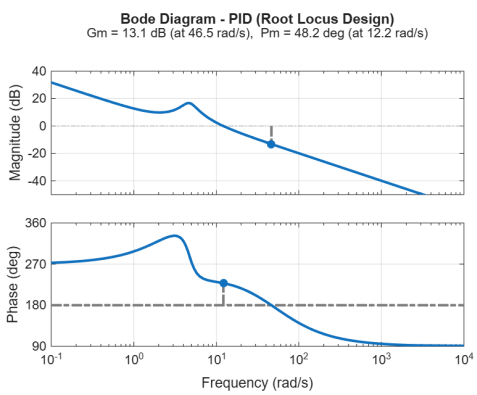

Fig. 18. Open-Loop Bode Diagram for PID (Frequency Response Design).

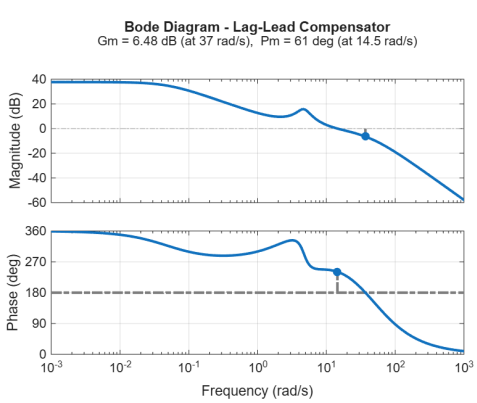

Fig. 19. Open-Loop Bode Diagram for Lag-Lead.

G. Summary of Final Transfer Functions & Performance

A comprehensive comparison of the final validated controllers is provided in Table VII. The Lag-Lead (Frequency Response) is the only design that strictly satisfies the settling time requirement (1.439 < 1.5 s) while maintaining a robust Phase Margin (> 60°).

H. Summary of Final Controllers

Following the design and iterative pole/zero placement processes described above, the final controller parameters selected for simulation and implementation are summarized in Table VI.

TABLE VI
FINAL ITERATION CONTROLLER PARAMETERS

Control Strategy	Parameters / Transfer Function
PID (Root Locus)	$K_p = 3$ $K_i = 4.85$ $K_d = 0.57$
PID (Freq. Response)	$K_p = 3.42$ $K_i = 5.13$ $K_d = 0.57$
Lag-Lead	$C(s) = 72\left(\frac{s+4}{s+38}\right)^2\left(\frac{s+6}{s+0.05}\right)$

It is clear that both design approaches lead to very similar PID gain values, whereas the Lag-Lead designs resulted in the exact same TF.

TABLE VII
FINAL PERFORMANCE SUMMARY

Design Units	PM deg	GM dB	BW rad/s	T_s s	OS %
PID (RL)	48.25	13.12	22.93	1.516	10.48
PID (Freq)	44.98	12.95	23.43	1.500	14.85
LL (RL & Freq)	61.03	6.48	41.58	1.439	10.96

VI. EXPERIMENTAL RESULTS

The control strategies designed and simulated in Section V were deployed on the physical 1-DOF copter hardware to validate their performance under real-world conditions. The experimental setup utilizes an ESP32 microcontroller for real-time data acquisition and control signal generation, interfacing with MATLAB/Simulink via serial communication.

A. Control Implementation

The final control structure implemented in Simulink is illustrated in Fig. 23. A critical addition to the theoretical design was the feed-forward bias voltage (1.1 V), which compensates for the gravitational offset identified in Section III (System Analysis). Furthermore, to mitigate the sensor noise a Low Pass Filter (LPF) was introduced in the feedback path.

B. Design Selection & Justification

Following the comparative analysis of simulation results and physical hardware testing, a strategic decision was made regarding the final controller selection.

While the Lag-Lead Compensator exhibited superior theoretical performance in simulation (higher bandwidth and Phase Margin), its implementation on the physical 1-DOF copter revealed practical limitations. The derivative action in the Lead network amplified high-frequency sensor noise, resulting in slow response to disturbance and potential motor saturation. This sensitivity to noise and unmodelled dynamics made the Lag-Lead design less suitable for the specific sensors used in this low-cost platform.

Conversely, the PID Controller demonstrated superior robustness in the presence of real-world noise and friction. The intuitive tuning structure allowed for the optimization of the derivative filter coefficient (N) to mitigate noise without compromising stability. Consequently, the PID controller was selected as the final production solution. It successfully met all critical performance specifications ($\pm 3°$ tracking error, $T_s < 1.5$ s, $M_p < 15\%$) while offering a more robust and reliable implementation for the hardware constraints defined by the client.

Despite the rigorous design process, discrepancies between the theoretical linear model and the physical plant such as unmodelled aerodynamic drag, non-linear Coulomb friction, and motor dead-zone are inevitable. Consequently, the analytically derived controller parameters serve as a near-optimal starting point rather than a final solution. To compensate for these unmodelled dynamics, manual fine-tuning is performed. This iterative process focused on ensuring the system meets the required performance targets across the entire operating region, beyond the limited linear range, while maintaining additional safety margins.

C. Performance Analysis

The closed-loop step response of the final tuned system is presented in Fig. 21. The system tracks the commanded reference angle of 15° with zero steady-state error, validating the integral action of the controller.

- Settling Time: The system achieves a settling time of 1.42 s (5% criterion), strictly satisfying the $T_s \leq 1.5$ s requirement.
- Overshoot: The maximum overshoot observed is approximately 12%, which is within the safety limit of 15%.
- Robustness: The controller successfully rejects disturbances caused by aerodynamic drag and minor friction non-linearities, maintaining a stable hover at the setpoint.

Table VIII presents the comparison between the theoretically derived PID parameters (from Section V) and the final experimentally tuned values.

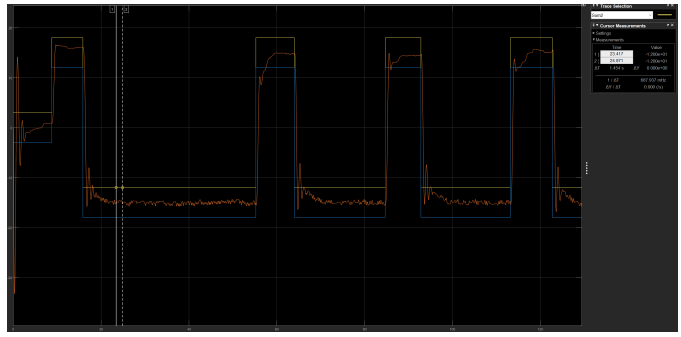

Fig. 20. Experimental square-wave response of the PID controlled system compared to simulation.

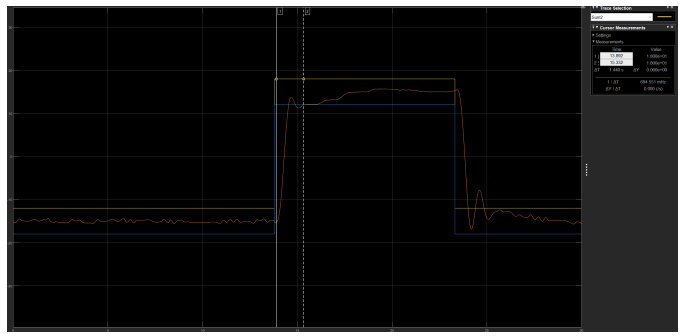

Fig. 21. Experimental step response of the PID controlled system compared to simulation.

VII. Conclusion

This project successfully achieved the design and implementation of a high-performance position controller for a 1-DOF copter system. A physics-based dynamic model was first derived and refined through experimental system identification, revealing a Type 0 plant behaviour dominated by gravitational stiffness. Three distinct control strategies were developed and analysed. While the Proportional controller failed to meet accuracy requirements and the Lag-Lead compensator exhibited sensitivity to sensor noise, the PID controller proved to be the optimal practical solution. The final design satisfies all technical specifications, including a settling time of 1.42 s (< 1.5 s), overshoot of 12% (< 15%), and zero steady-state error, while strictly adhering to the cost (< \$500) and timeline constraints. Future work may investigate Full-State Feedback control using an Observer to further improve disturbance rejection, or the implementation of an adaptive control scheme to automatically compensate for changing payload masses. The delivered system

TABLE VIII
Comparison of Initial Design vs. Final Tuned PID Parameters

Parameter	Initial Design	Final Tuned (Exp)
K_p	3.00	1.48
K_i	4.85	1.25
K_d	0.57	0.48

stands as a robust, compliant, and cost-effective platform ready for deployment in academic laboratories.

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VIII. Nomenclature

TF	Transfer Function
RL	Root Locus
FR	Frequency Response
OLTF	Open-Loop Transfer Function
CLTF	Closed-Loop Transfer Function
PID	Proportional–Integral–Derivative
LL	Lag–Lead

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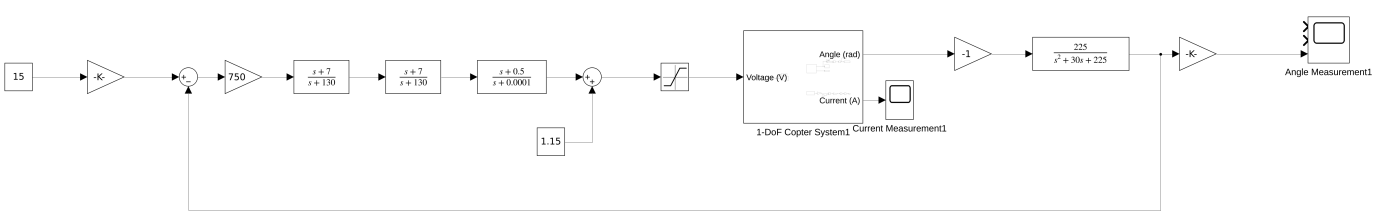

Fig. 22. Simulink circuit for the experimental results of the final tuned Lag-Lead controller.

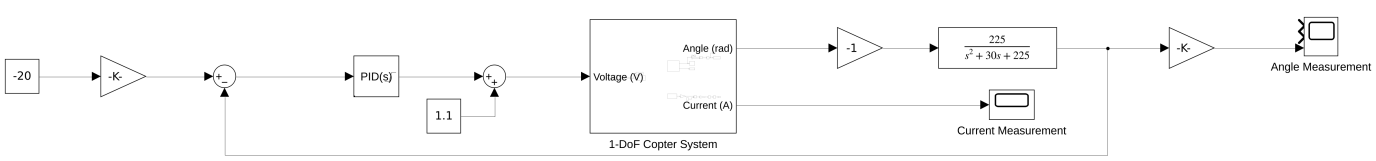

Fig. 23. Simulink circuit for the experimental results of the final tuned PID controller.